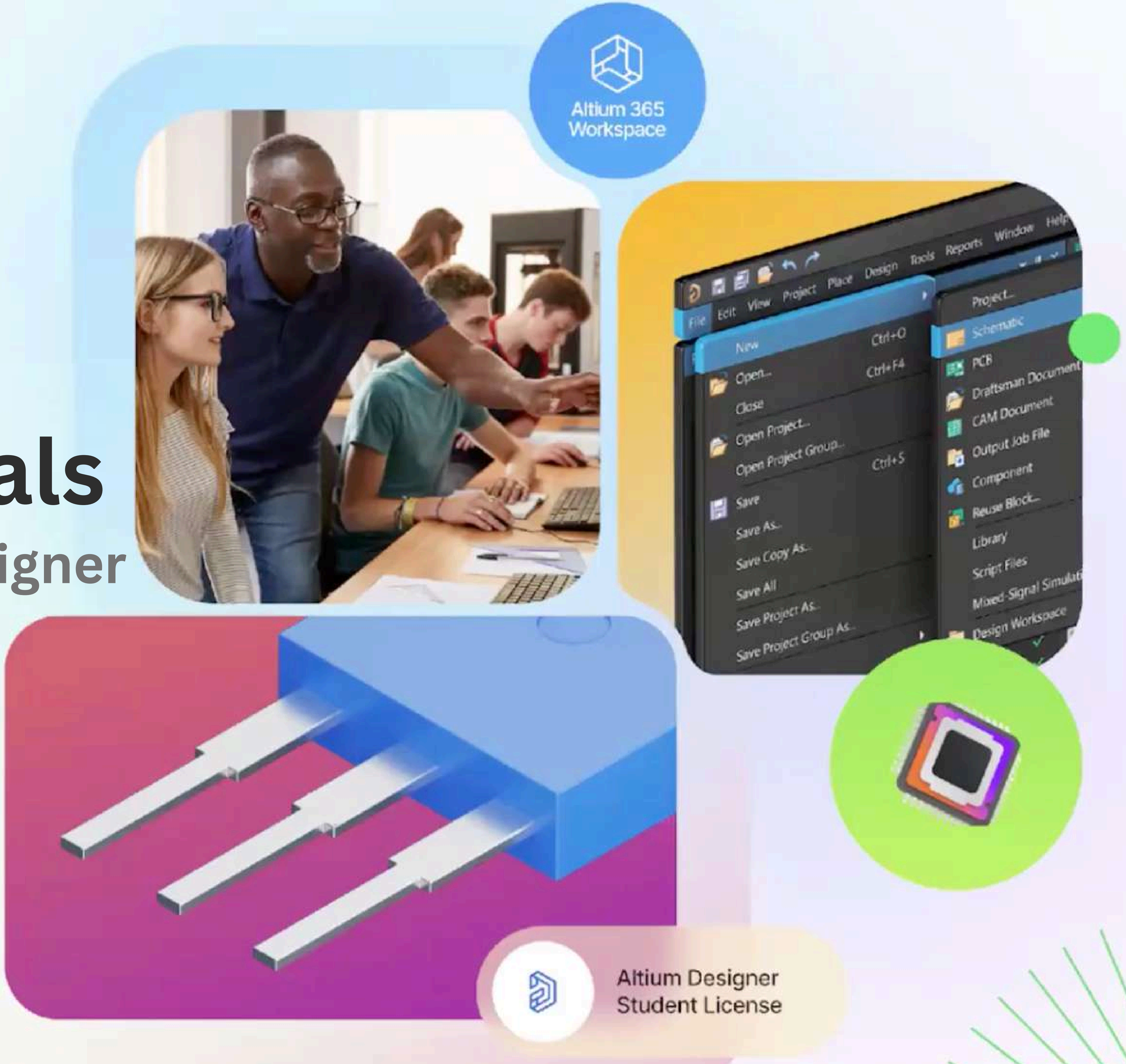
Altium EDUCATION

IIT GOA
INDIAN INSTITUTE OF TECHNOLOGY GOA
भारतीय प्रौद्योगिकी संस्थान गोवा

3-Day Bootcamp

PCB Design Fundamentals

Hands-on using Altium Designer



Praveen Saxena

Altium Global Ambassador, IIT Goa

About Me

Praveen Saxena

Altium Global Ambassador

M.Tech – VLSI & Signal Processing, IIT Goa

Embedded systems and PCB design enthusiast with hands-on experience in real-time DSP, IoT systems, and sensor-based product development.

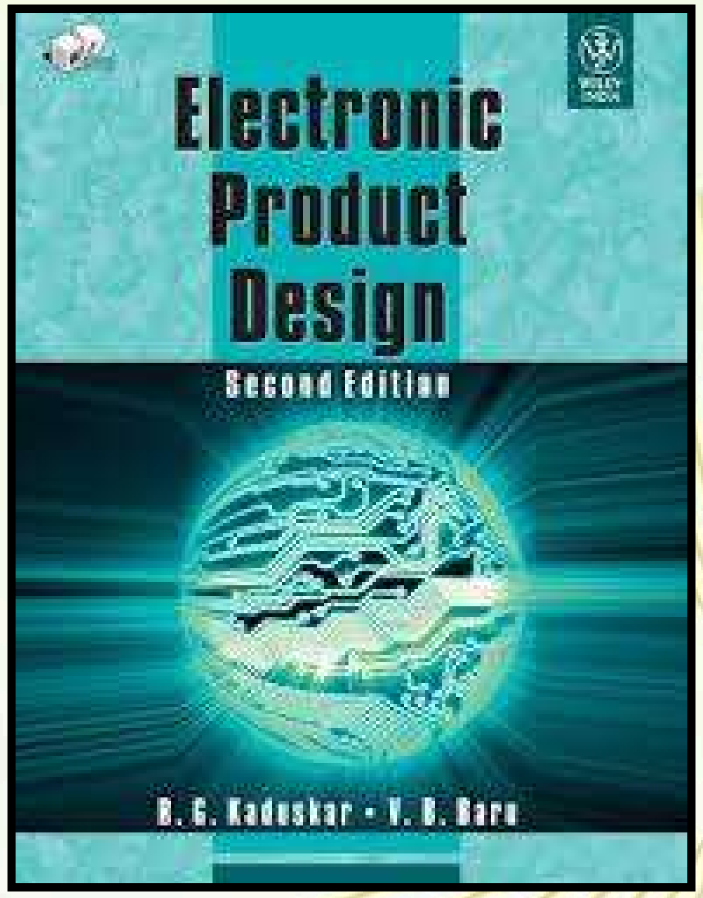
- Teaching Assistant – IoT & Digital Communication Labs (IIT Goa)
- Working on Smart Periodontal Probe (sensor-based medical device)
- Experience with Drone, ESP32, Raspberry Pi & real-time systems

Training agenda

- Introduction to PCB Design
- Schematic to Layout Workflow
- Hands-on Practice
- Industry-Level Design Concepts

Reference Book

- Electronic Product Design – R. G. Kadus



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Day 1 Objectives

- Basics of PCB & its structure
- Schematic vs PCB understanding
- Components & mounting technologies
- Breadboard vs PCB comparison
- Limitations of manual wiring
- Introduction to Altium Designer
- Overview of PCB design workflow

Day 2 Objectives

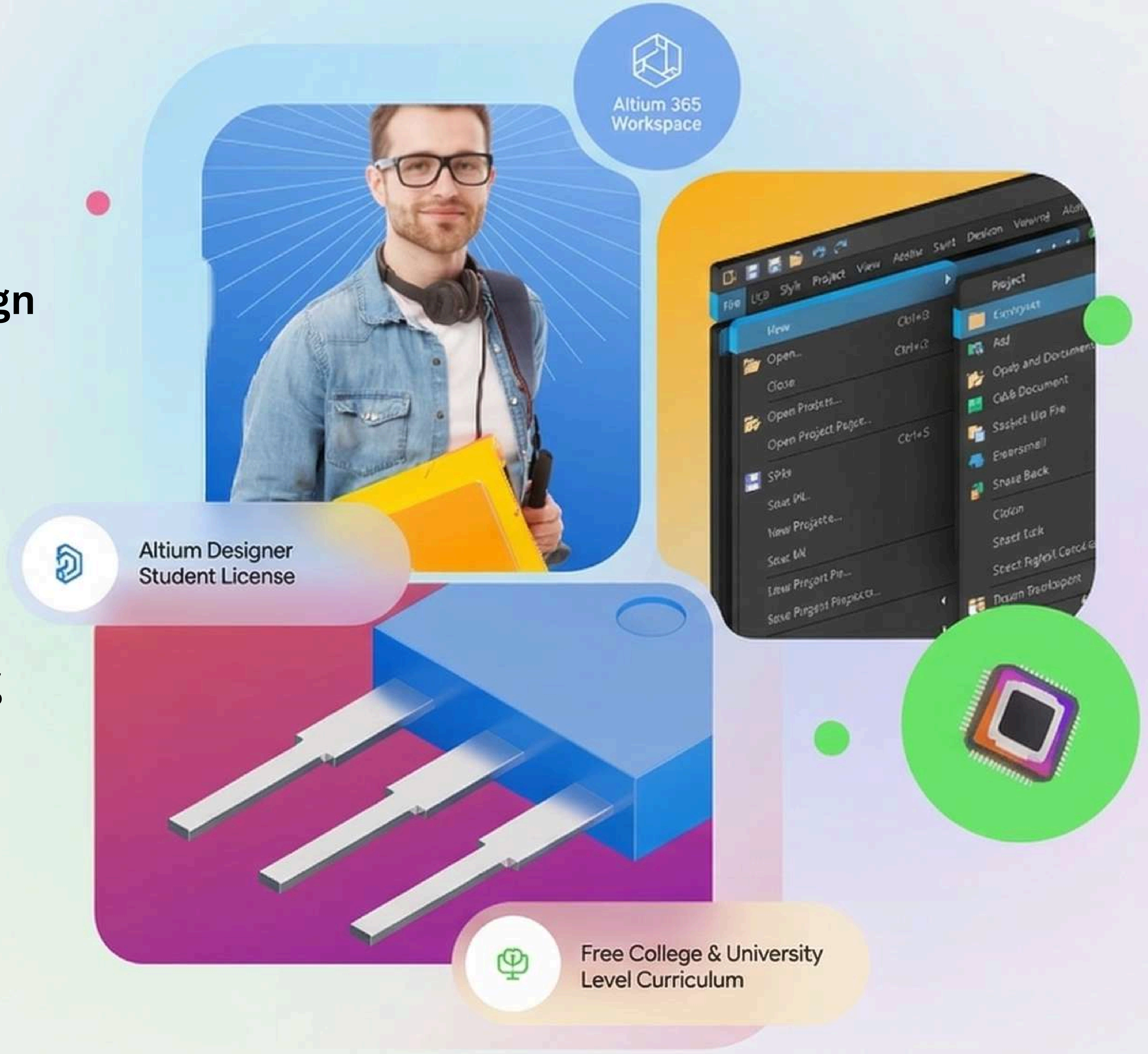
- Install Altium Designer
- Set up workspace and preferences
- Configure units and libraries
- Create a new project
- Understand schematic editor basics
- Design voltage regulator circuit (7805)
- Place components and make connections
- Compile and check errors
- Convert schematic to PCB layout
- Component placement (layout basics)
- Routing tracks (manual/auto)
- Run design rule check (DRC)
- Verify complete PCB layout

🎓 Certification Criteria

To earn the Certificate of Completion (PCB Design using Altium Designer):

📌 Requirements
✔ Attend the complete 3-day workshop
✔ Complete Altium Education free courses (Register at: https://education.altium.com/)
✔ Actively follow the course side-by-side during training
✔ Pass the final quiz assessment

Enroll for Free



CERTIFICATE OF COMPLETION

Certificate of Completion

Presented to

Praveen Saxena

FOR COMPLETION OF THE ALTIUM EDUCATION
PCB BASIC DESIGN COURSE

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PCB..?

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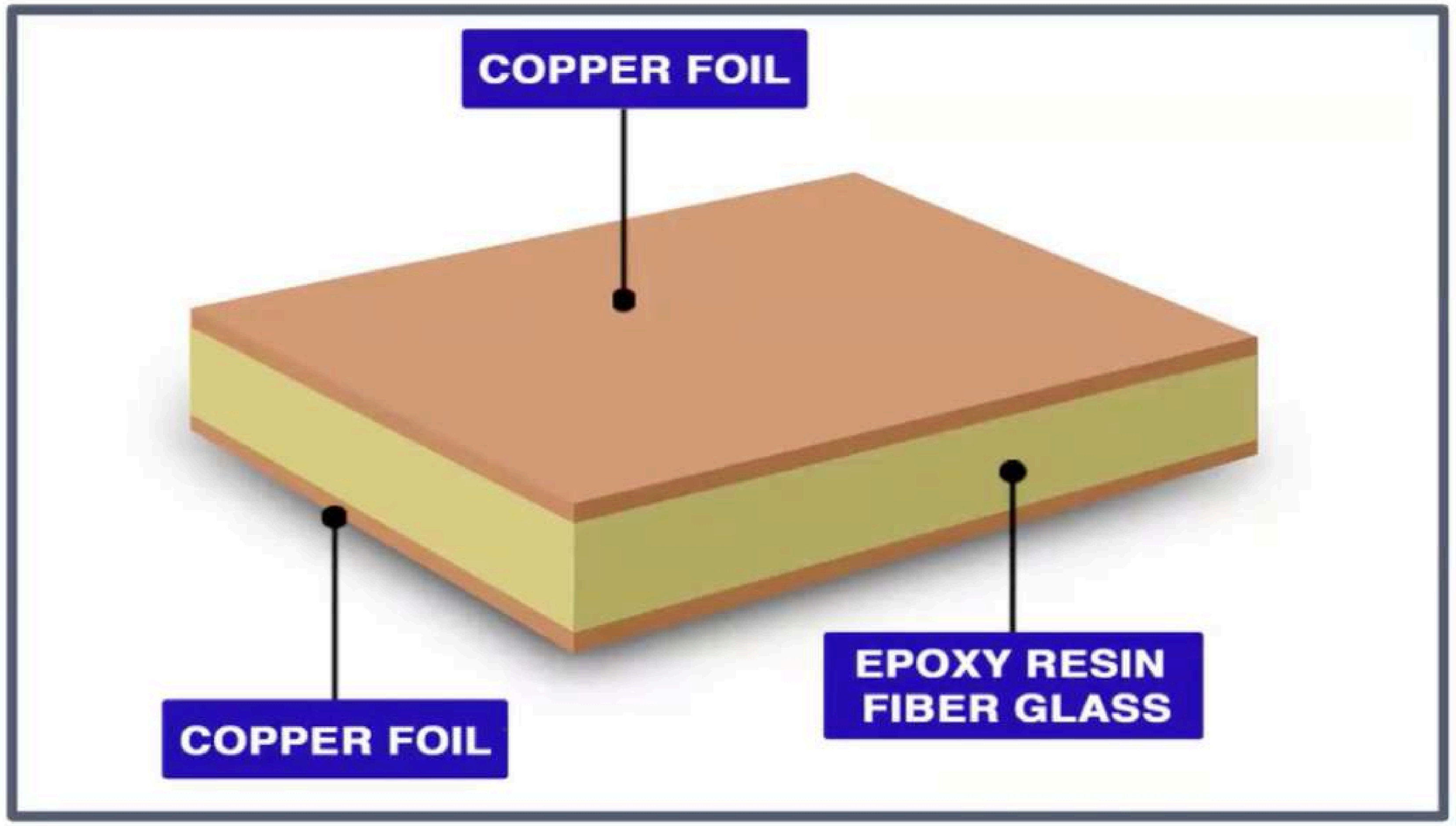
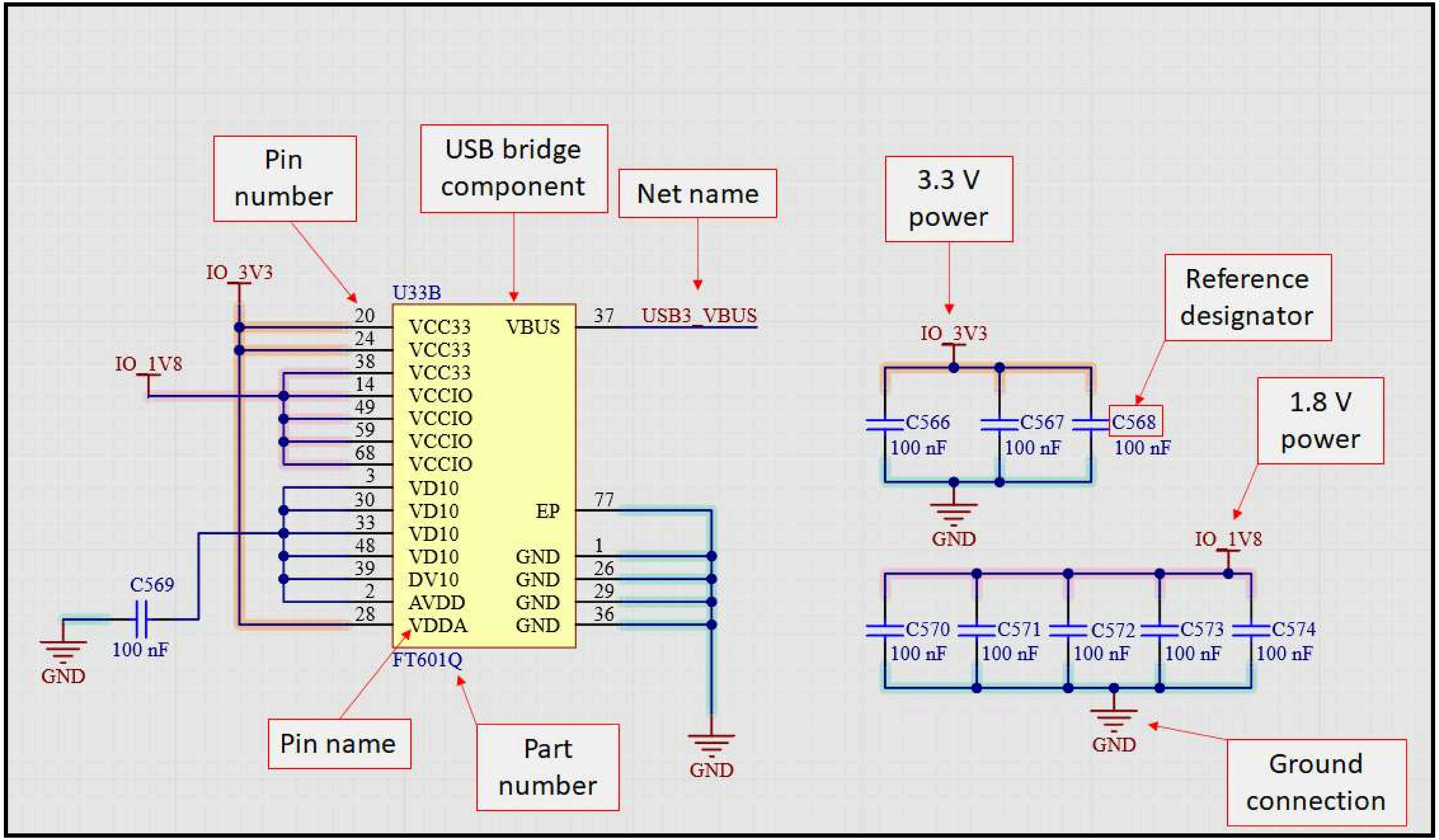


PCB = Mechanical support + Electrical interconnection

Schematic



Source: Altium Designer (Education)

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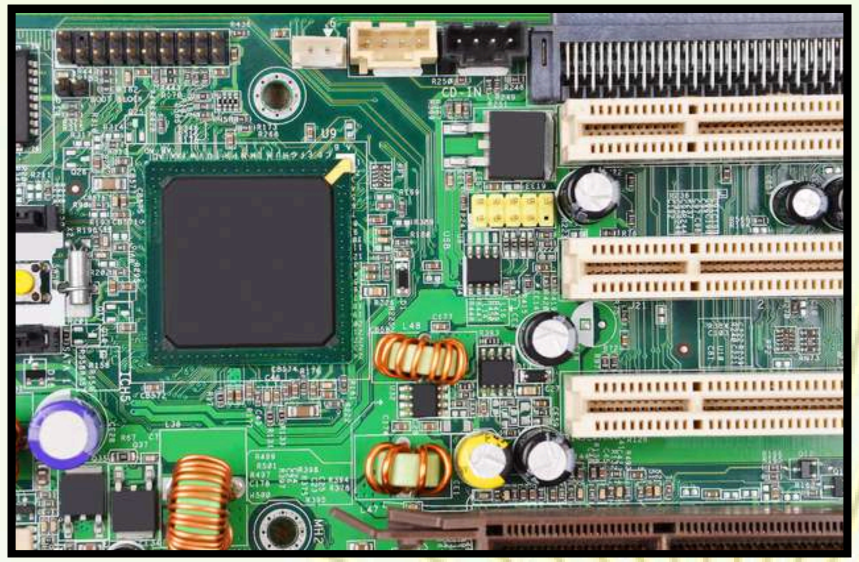
Introduction to PCB & PCB Design

PCB (Printed Circuit Board):

- Mechanically supports and electrically connects components
- Uses copper tracks on insulating substrate (FR4)
- Also called Printed Wiring Board (PWB)
- PCB with components = PCB Assembly



Importance:

- Final stage of circuit design
- First step toward product manufacturing
- Affects performance, reliability, and cost

Introduction to PCB & PCB Design

PCB Design:

- Process of converting schematic into physical layout
- Involves component placement and routing
- Ensures proper connectivity and manufacturability
- Can be simple or complex multilayer designs

Goal:

Schematic → PCB → Real Product

Common PCB Design Files

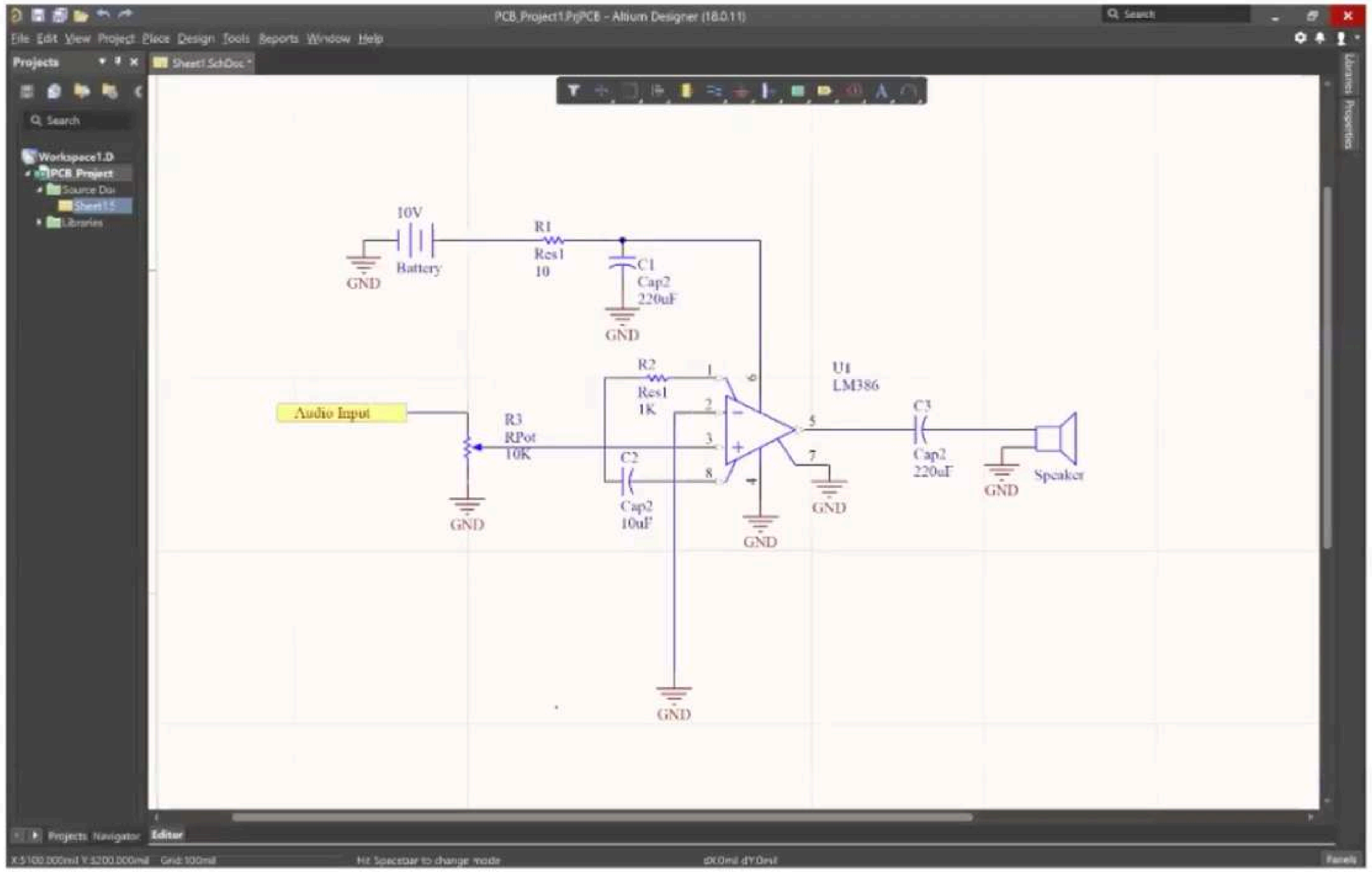
1. Schematic Sheets
- Electrical blueprint of circuit
- Shows components and connections

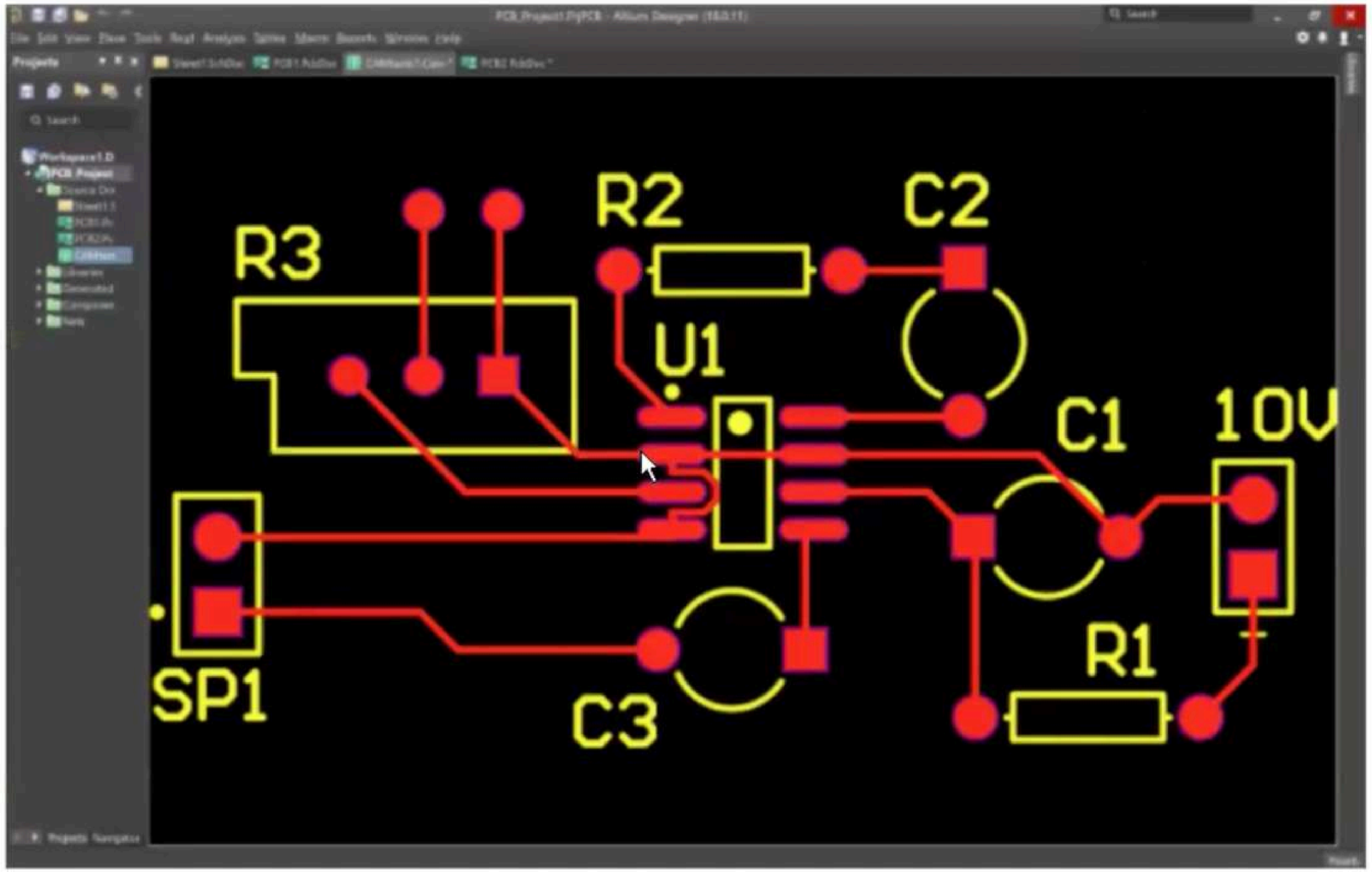
2. Bill of Materials (BOM)
- List of all components
- Used for ordering and assembly

3. PCB Layout
- Physical placement + routing
- Defines actual board structure

4. PCB Libraries
- Stores symbols, footprints, 3D models
- Includes component data (MPN, suppliers)

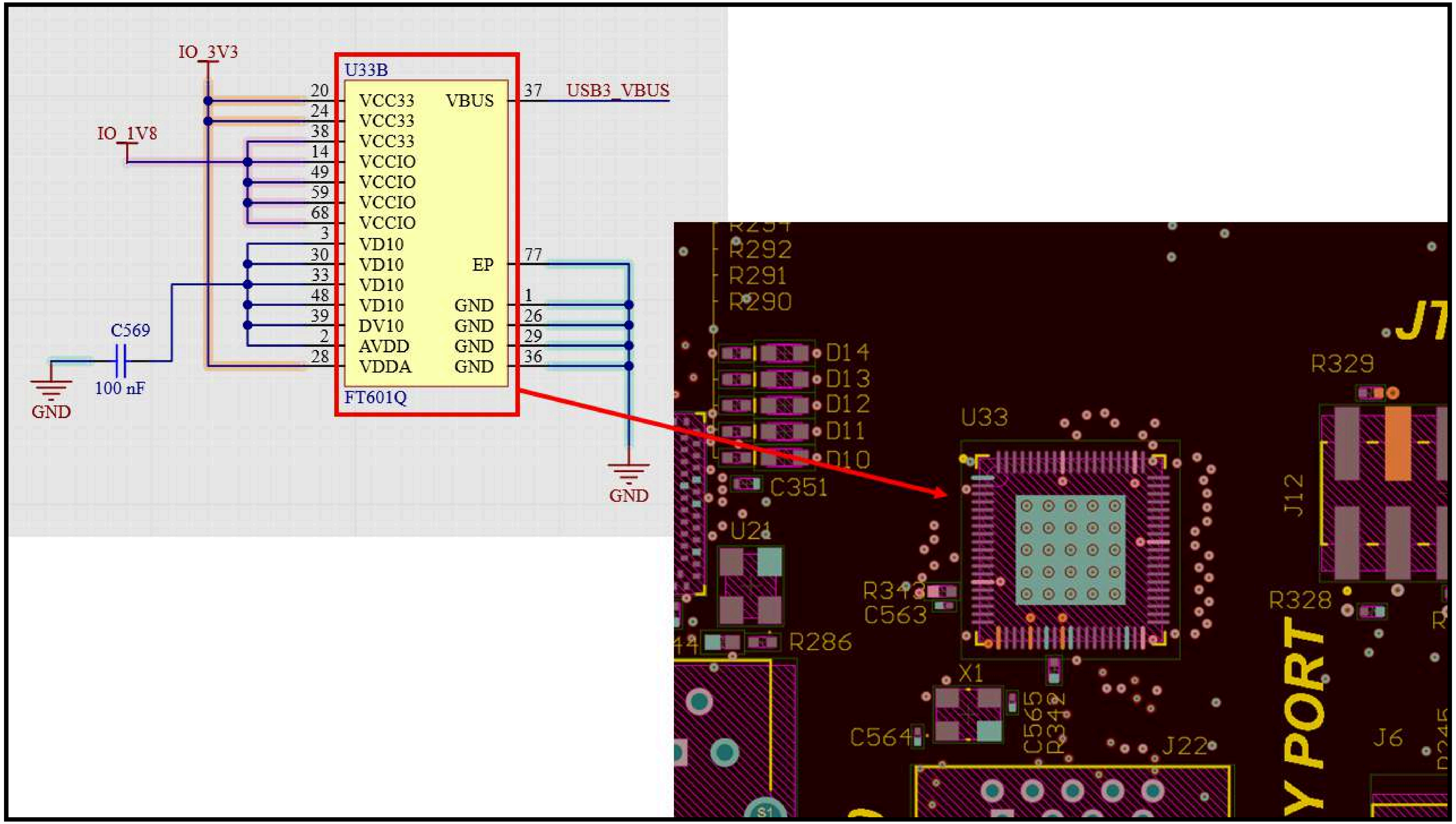
Schematic vs PCB





Source: Altium Designer (Education)

Schematic vs PCB



Source: Altium Designer (Education)

Important PCB Terms

Basic Terms:

- Component → Device mounted on PCB (R, C, IC)
- PCB → Bare board without components
- PCB Assembly (PCBA) → PCB with components
- Schematic → Logical circuit diagram
- PCB Layout → Physical design of board

Electrical & Layout:

- Trace (Track) → Copper connection
- Pad → Soldering point for components
- Via → Connection between layers
- Net → Electrical connection group

Important PCB Terms

Layers & Structure:

- Layers → Signal + Plane layers
- Multilayer PCB → >2 copper layers
- PCB Stackup → Arrangement of layers
- Plane Layer → Power/GND copper plane
- Signal Layer → Signal routing

Materials:

- Laminate → Insulating base (FR4)
- Copper-Clad Laminate → Laminate + copper
- Prepreg → Bonding resin layer

Important PCB Terms

Manufacturing:

- Plating → Metal coating
- Etching → Removing unwanted copper
- Photoresist → Light-sensitive coating
- Solder Mask → Protective coating

Key Idea:

These terms define PCB design, structure, and manufacturing

Types of PCBs:

PCBs are classified based on:

1. Material Used
2. Number of Layers

Types of PCBs:

1. Based on Material

- Conducting Material → Copper
- Dielectric Material → FR4, FR2, FR3, PTFE, Polyamide

Table 3.1 Types and specifications of boards

Board types	Specifications
Standard FR-4 Glass/Epoxy	Chemical resistant
High Performance FR-4	High flexural strength
FR-2 Paper/Phenolic	Flame resistant
FR-3 Paper/Epoxy	High mechanical strength
XXXPC Paper/Phenolic	Moisture resistant
Glass/Melanin	High impact strength
Polyamide Glass/Polyamide	High dimensional stability

Source: R. G. Kadus – Electronic Product Design

Types of PCBs:

2. Based on Number of Layers

(a) Single Layer PCB

- One copper layer
- Components on one side
- Uses through-hole or SMD
- Easy to design, debug, repair

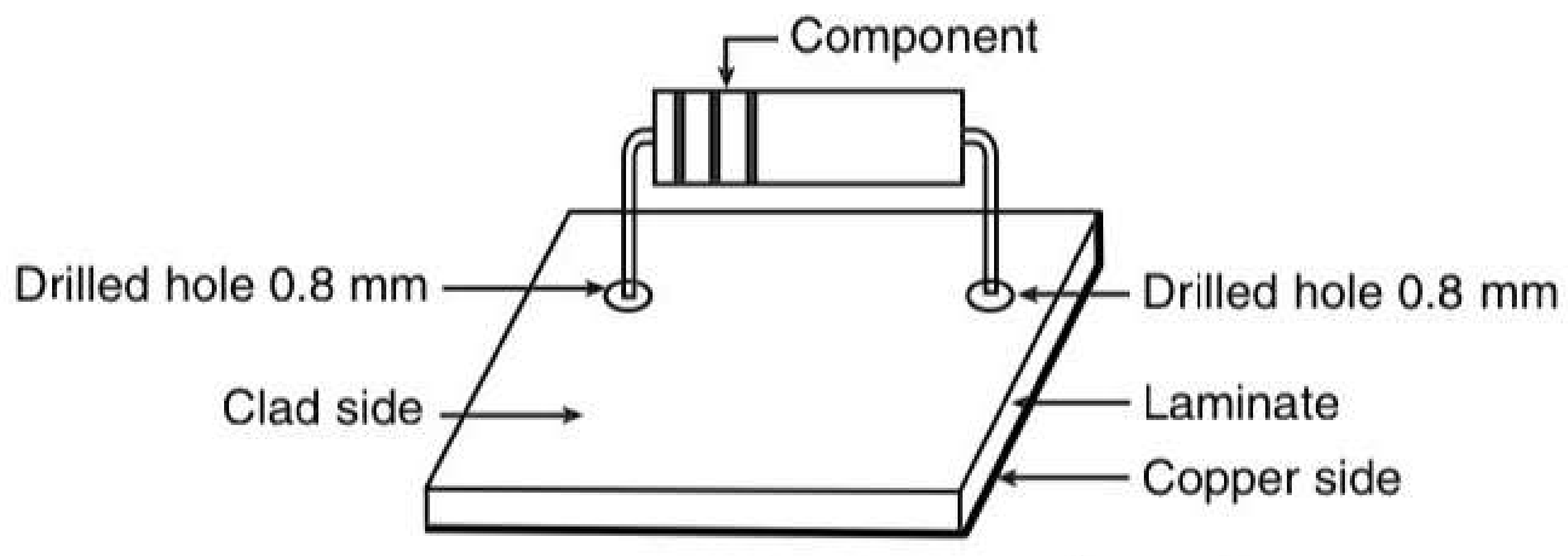


Figure 3.3 Component placed on a single-sided PCB.

Source: R. G. Kadus – Electronic Product Design

Types of PCBs:

2. Based on Number of Layers

(b) Double Layer PCB

- Copper on both sides
- Connection using vias (PTH)
- Components can be on both sides
- Higher density than single layer

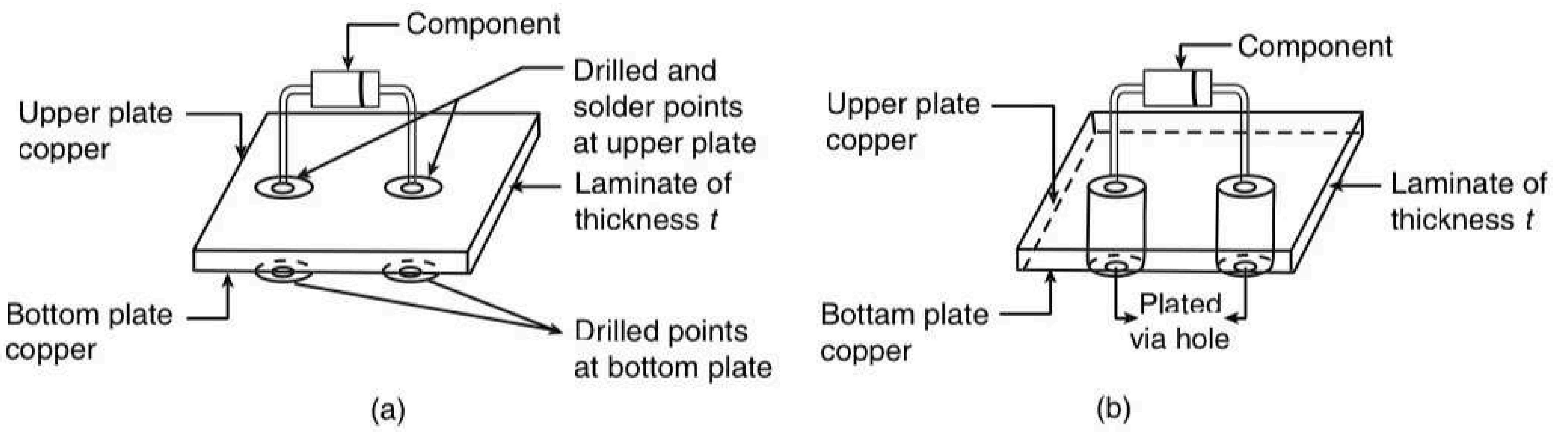


Figure 3.4 Double-sided PCB: (a) without PTH; (b) with PTH.

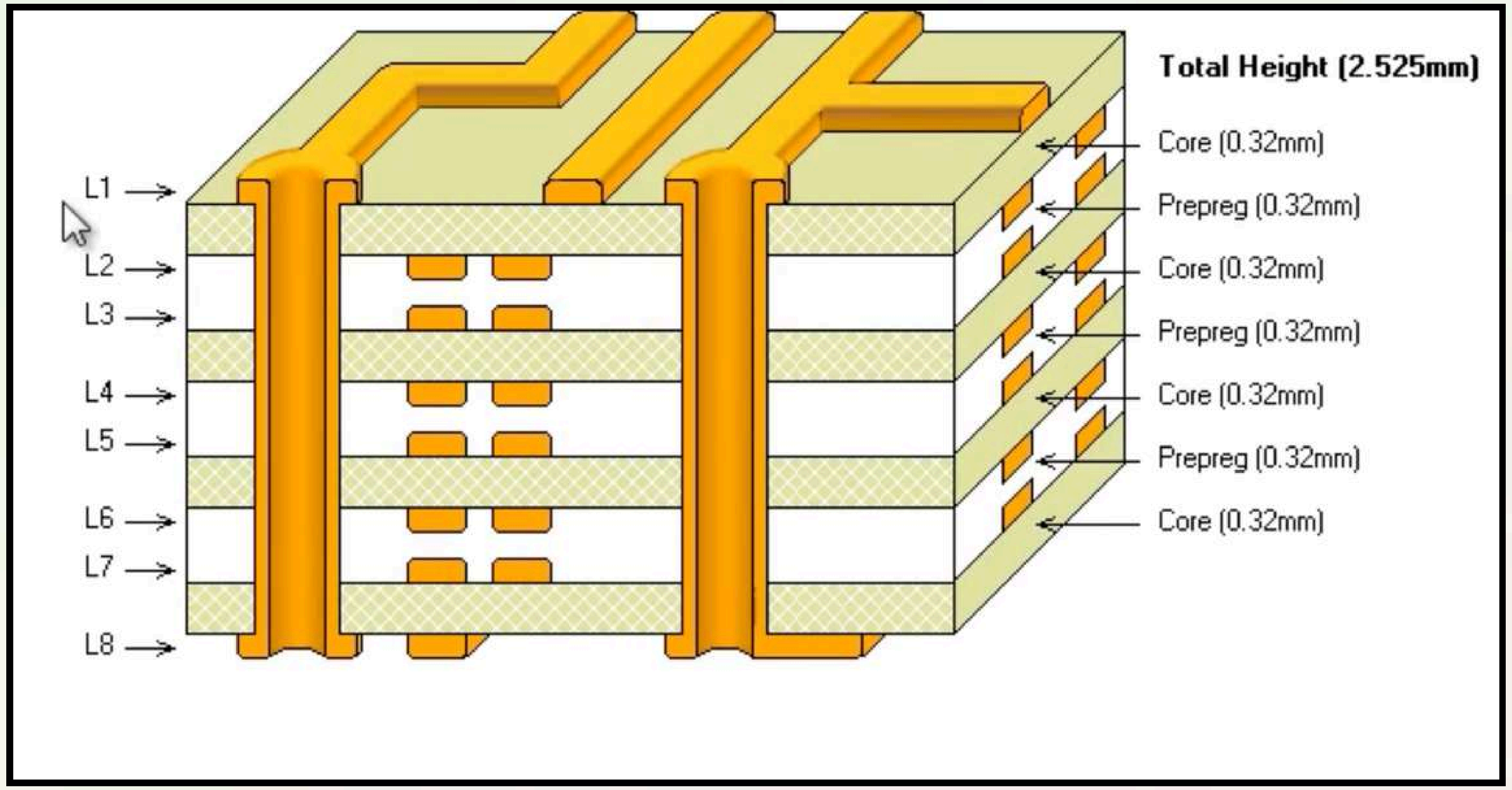
Source: R. G. Kadus – Electronic Product Design

Types of PCBs:

2. Based on Number of Layers

(c) Multilayer PCB

- More than two layers
- Sandwich structure
- Used for complex circuits
- Better routing and power distribution
- Higher cost



Source: Altium Designer (Education)

Mounting Technologies

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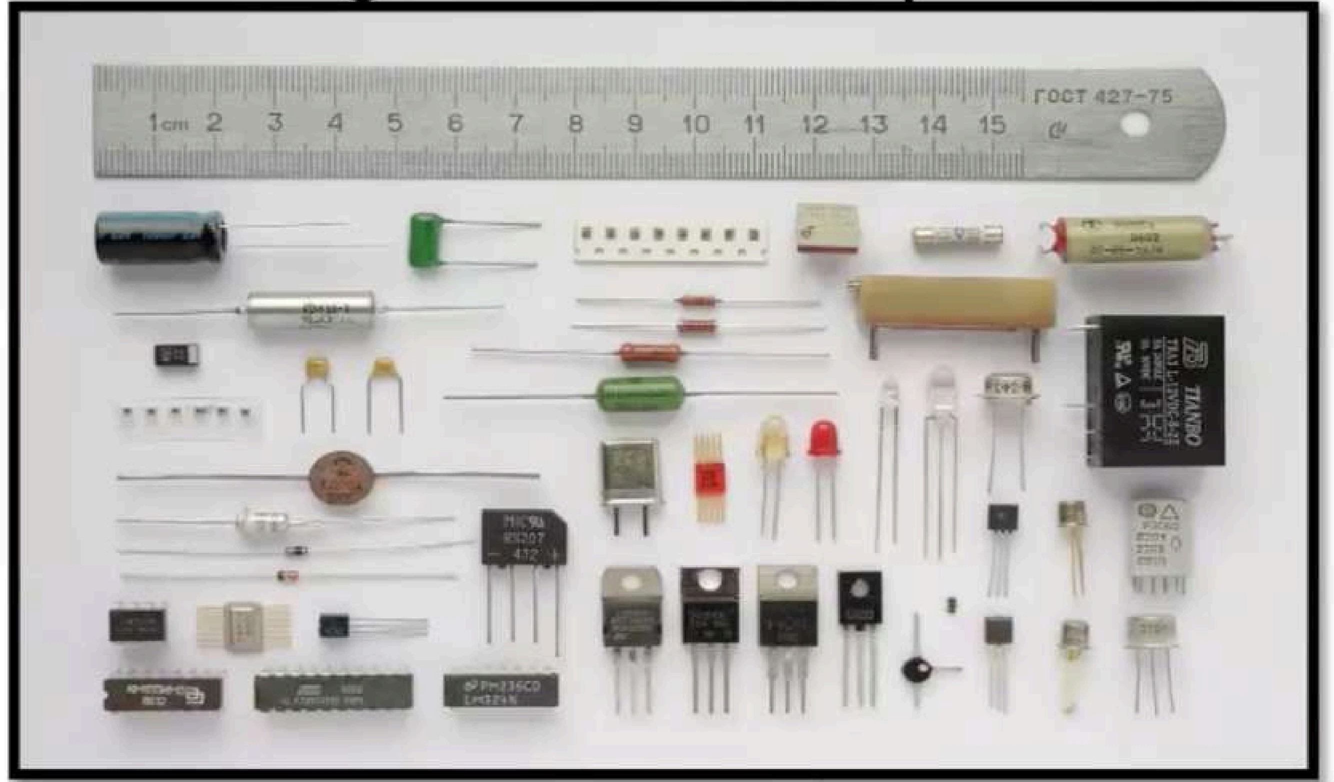
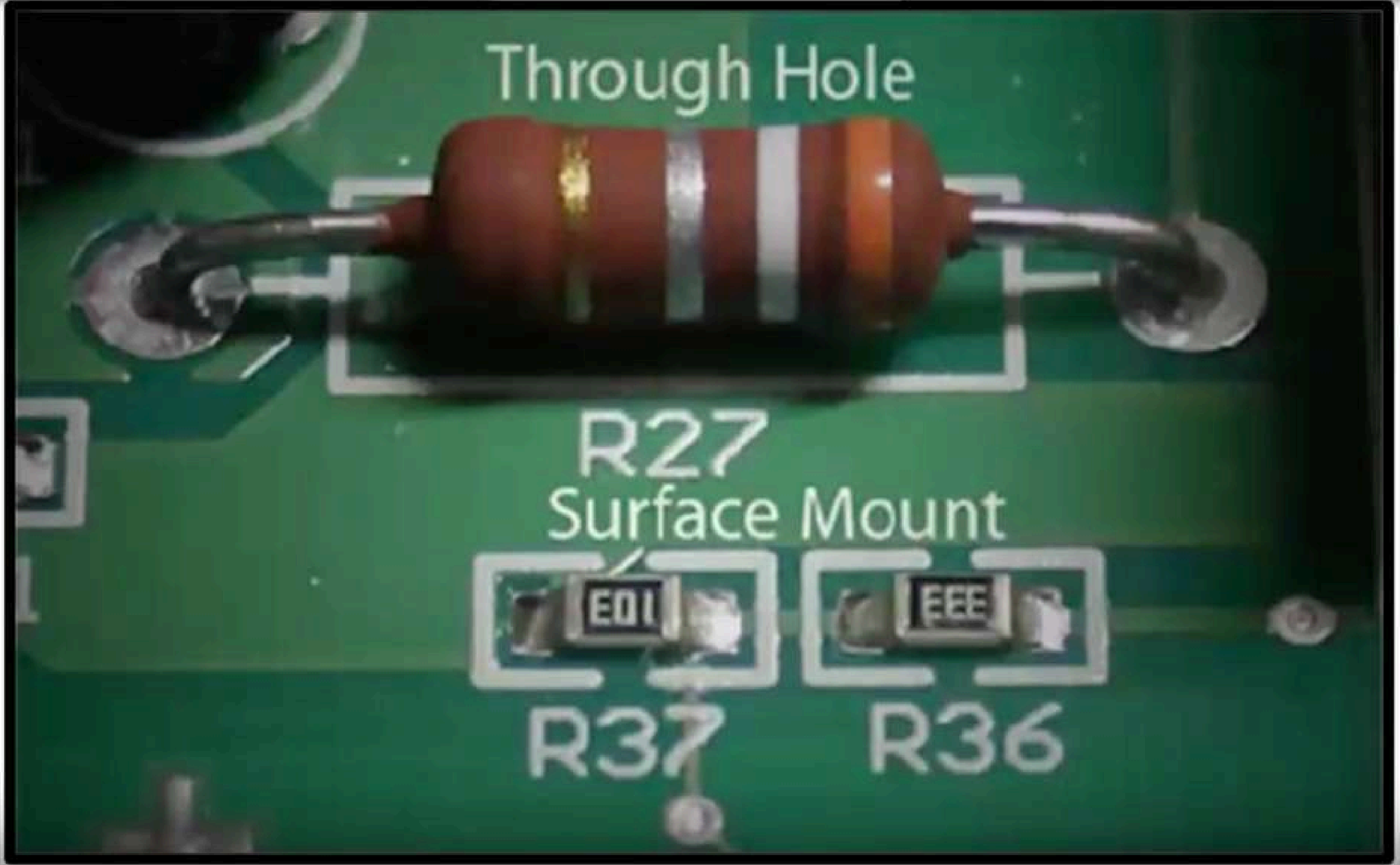
Through Hole Components



SMD (SMT) Components



Manually Drawn Logic Diagrams

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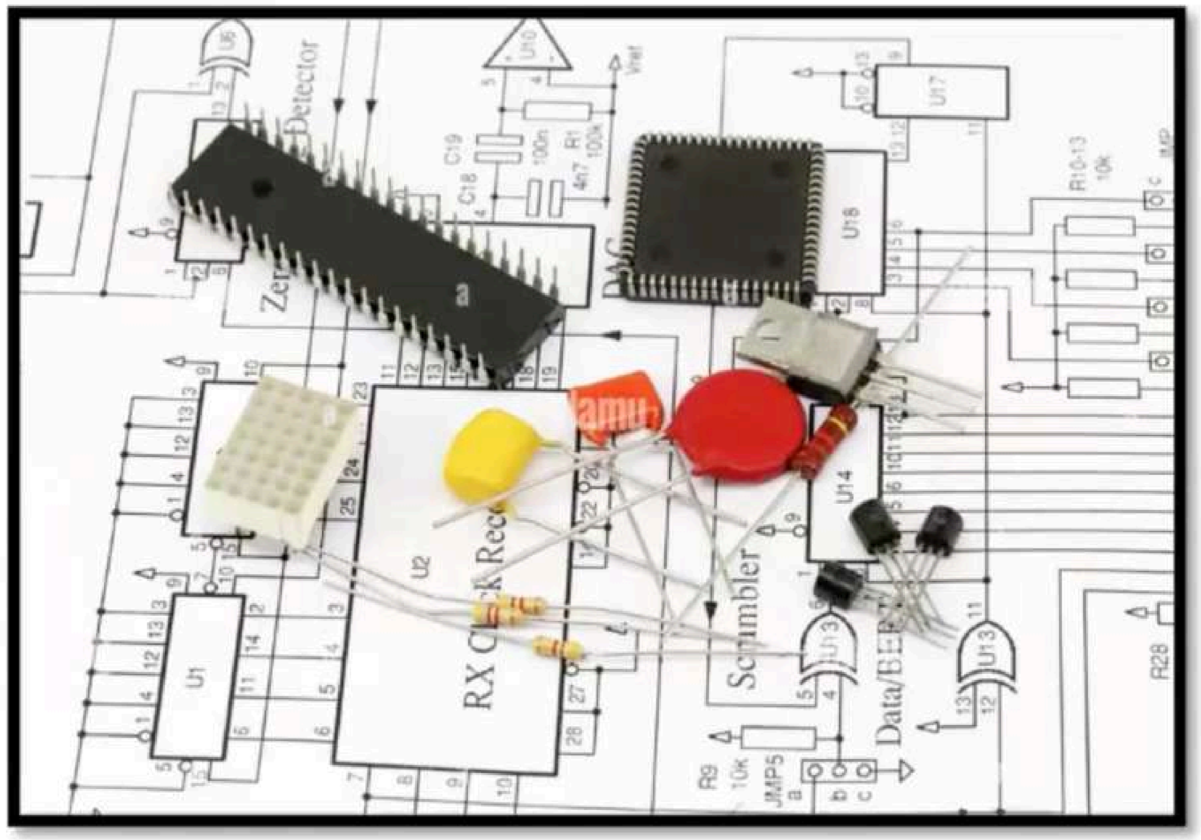
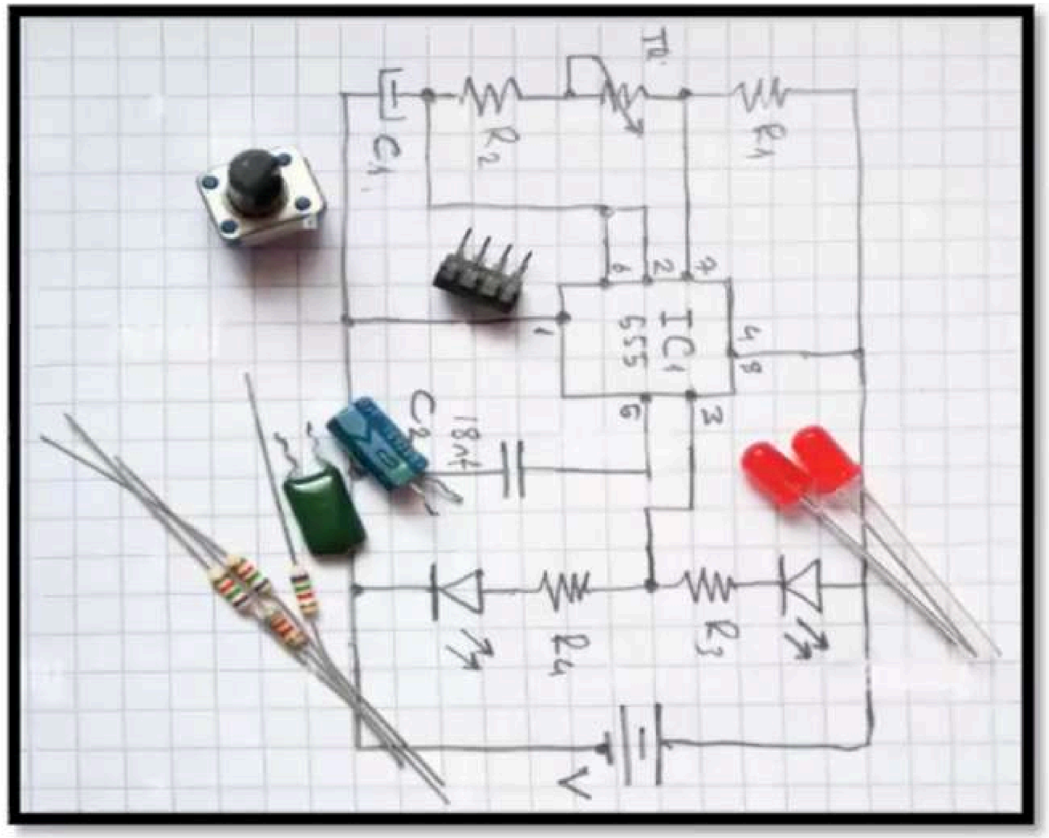
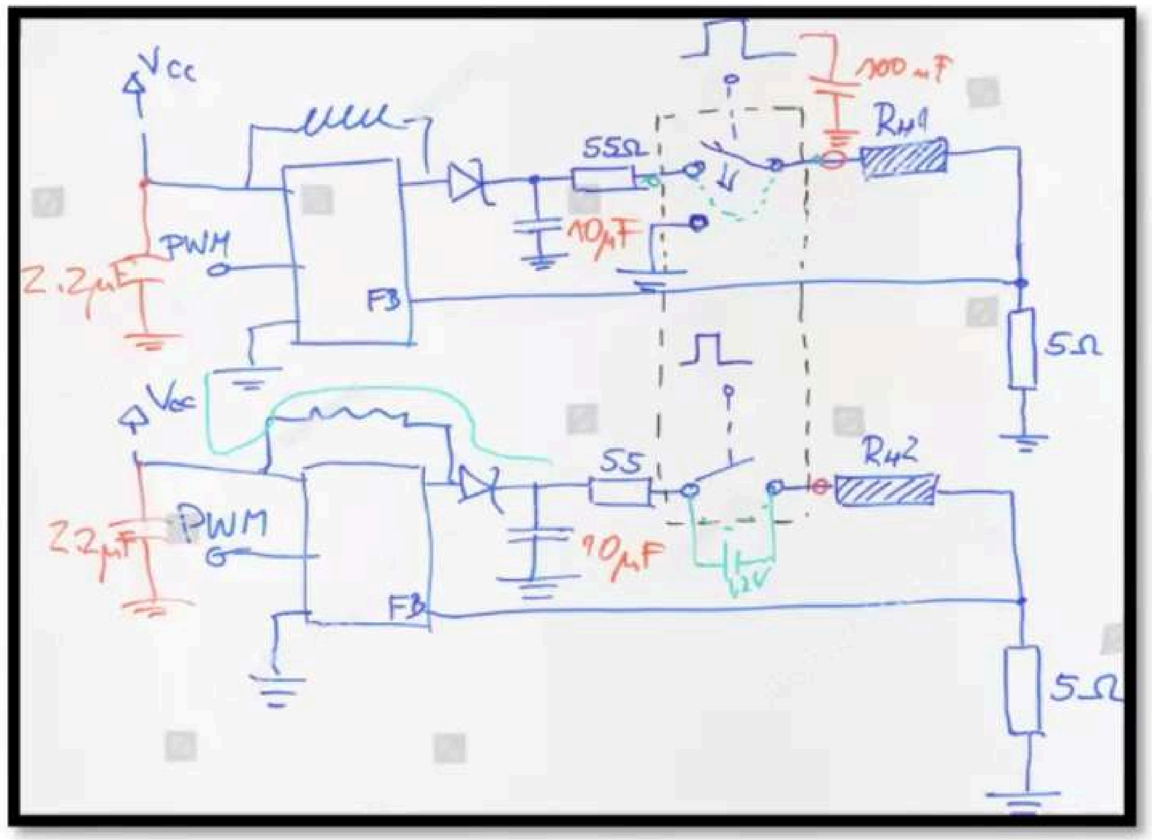
Free Handwritten Schematics
Stencil drafted Drawings

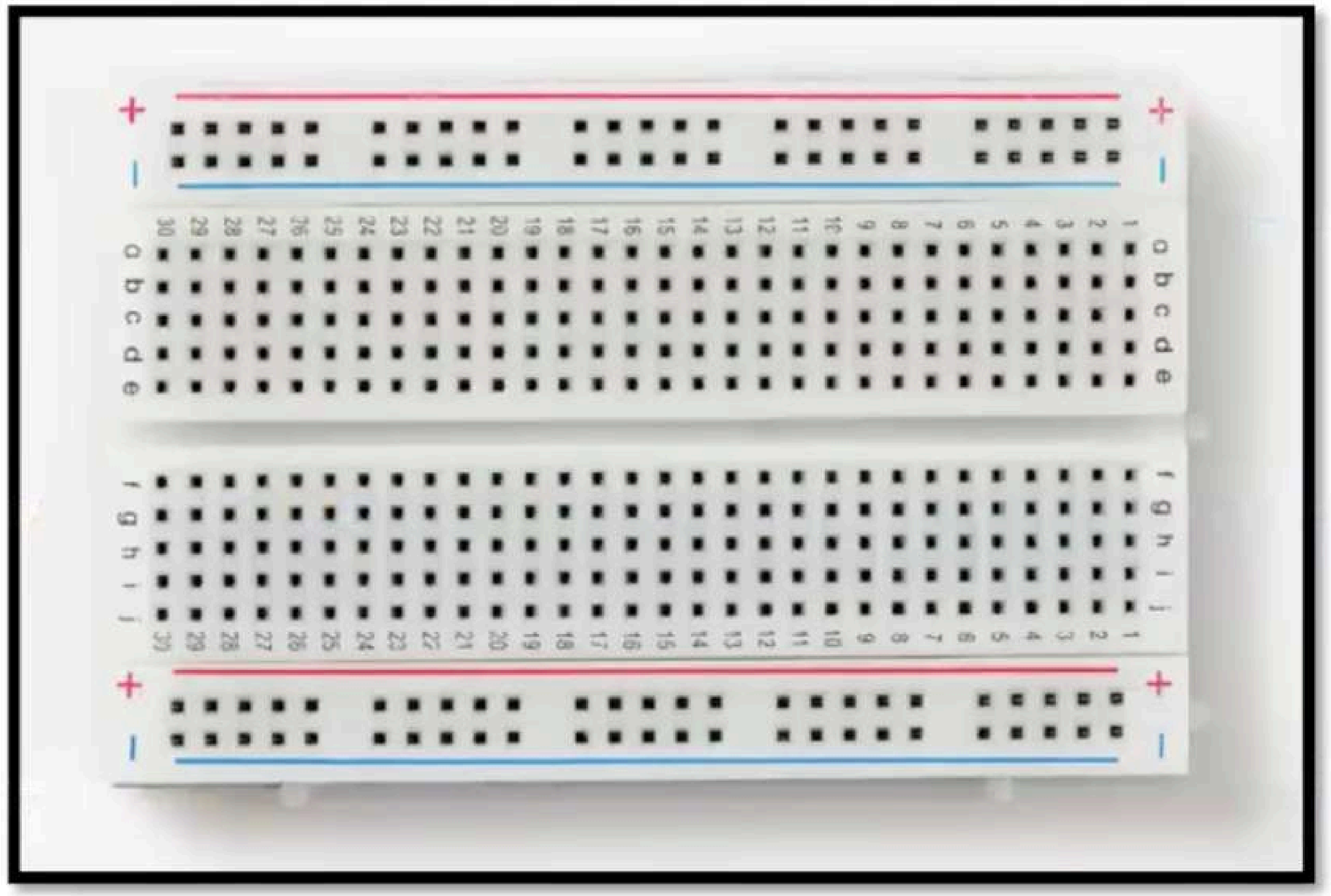




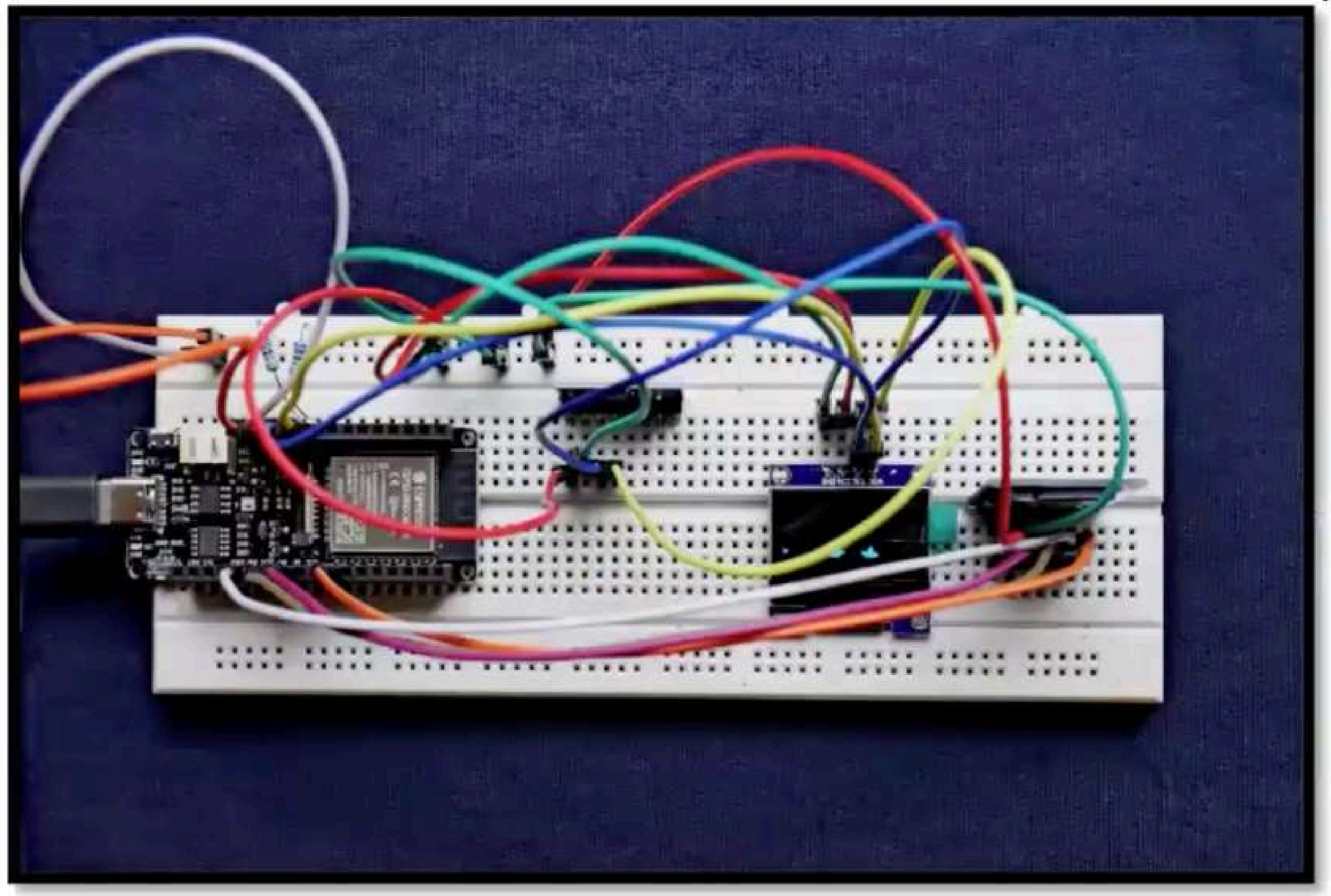


Building Circuits using Breadboards

Bare Breadboard

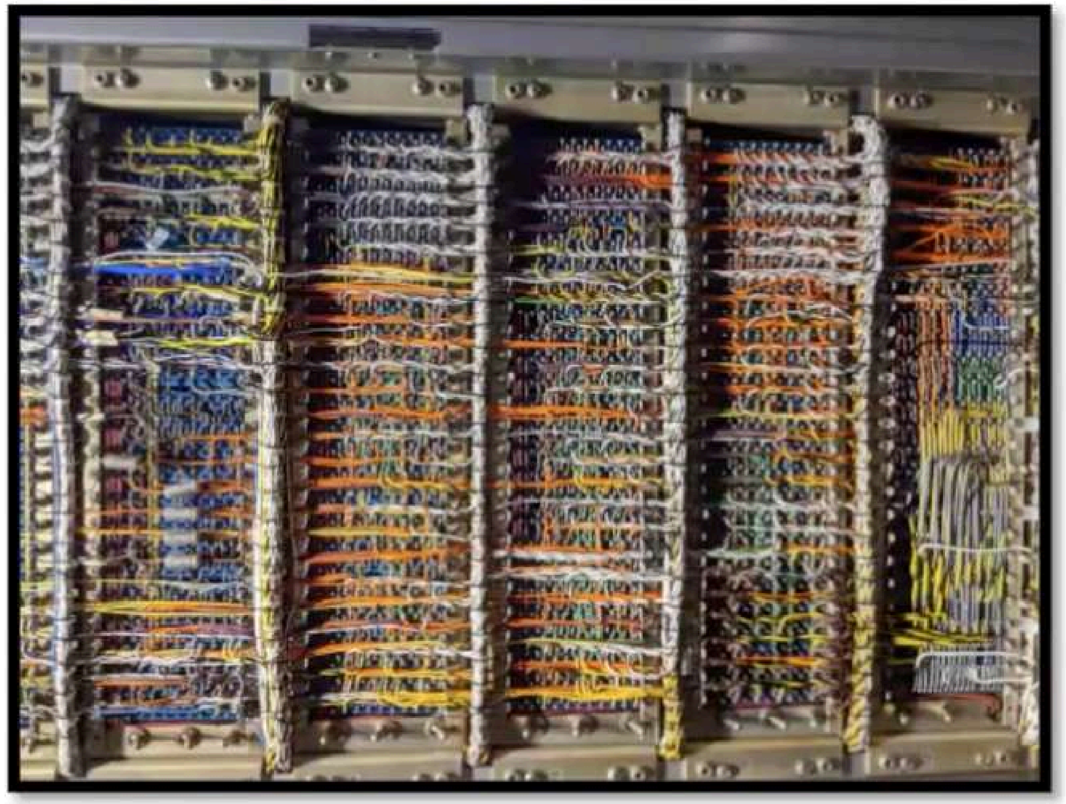


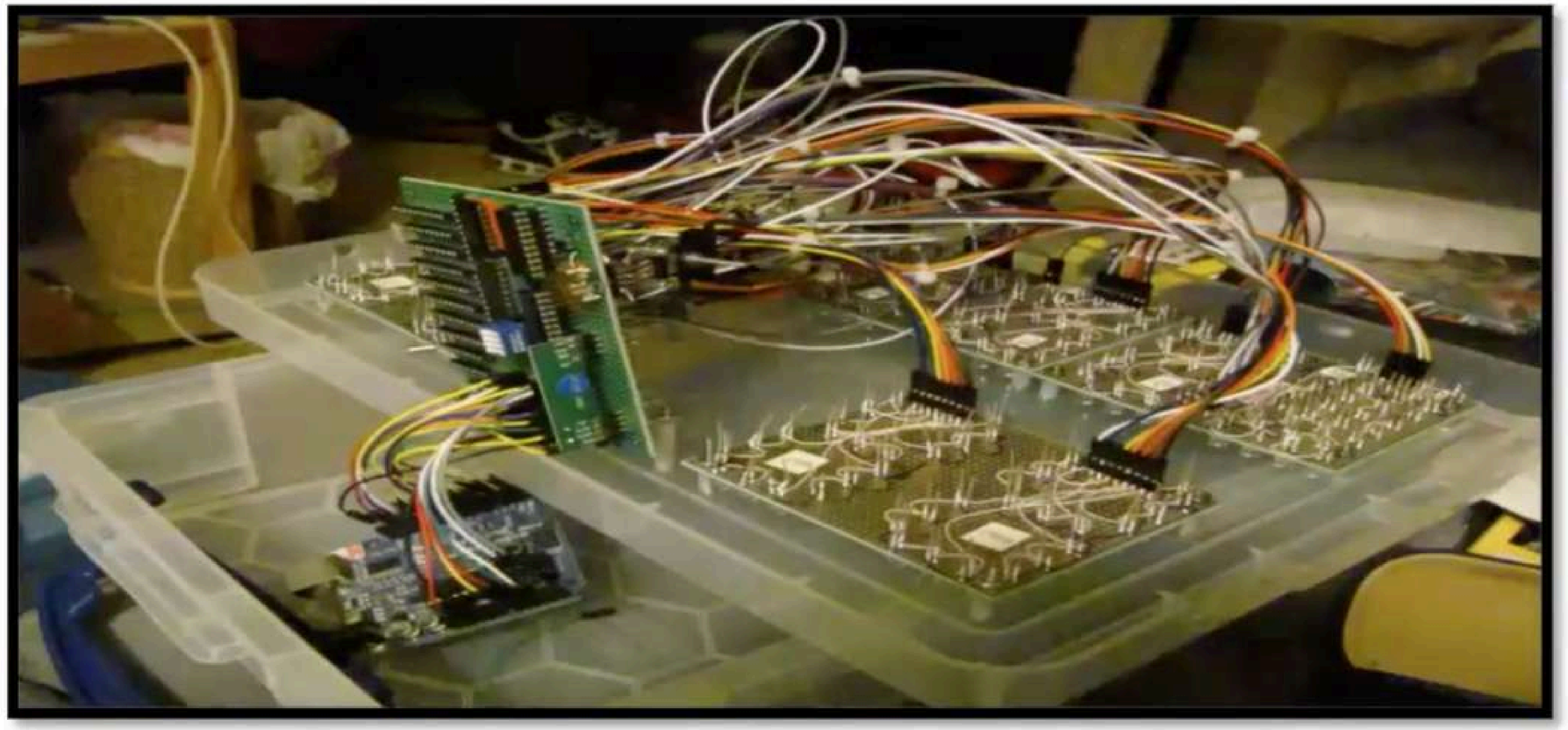
Populated and physically Wired Board



Challenges in Manually Wired Boards

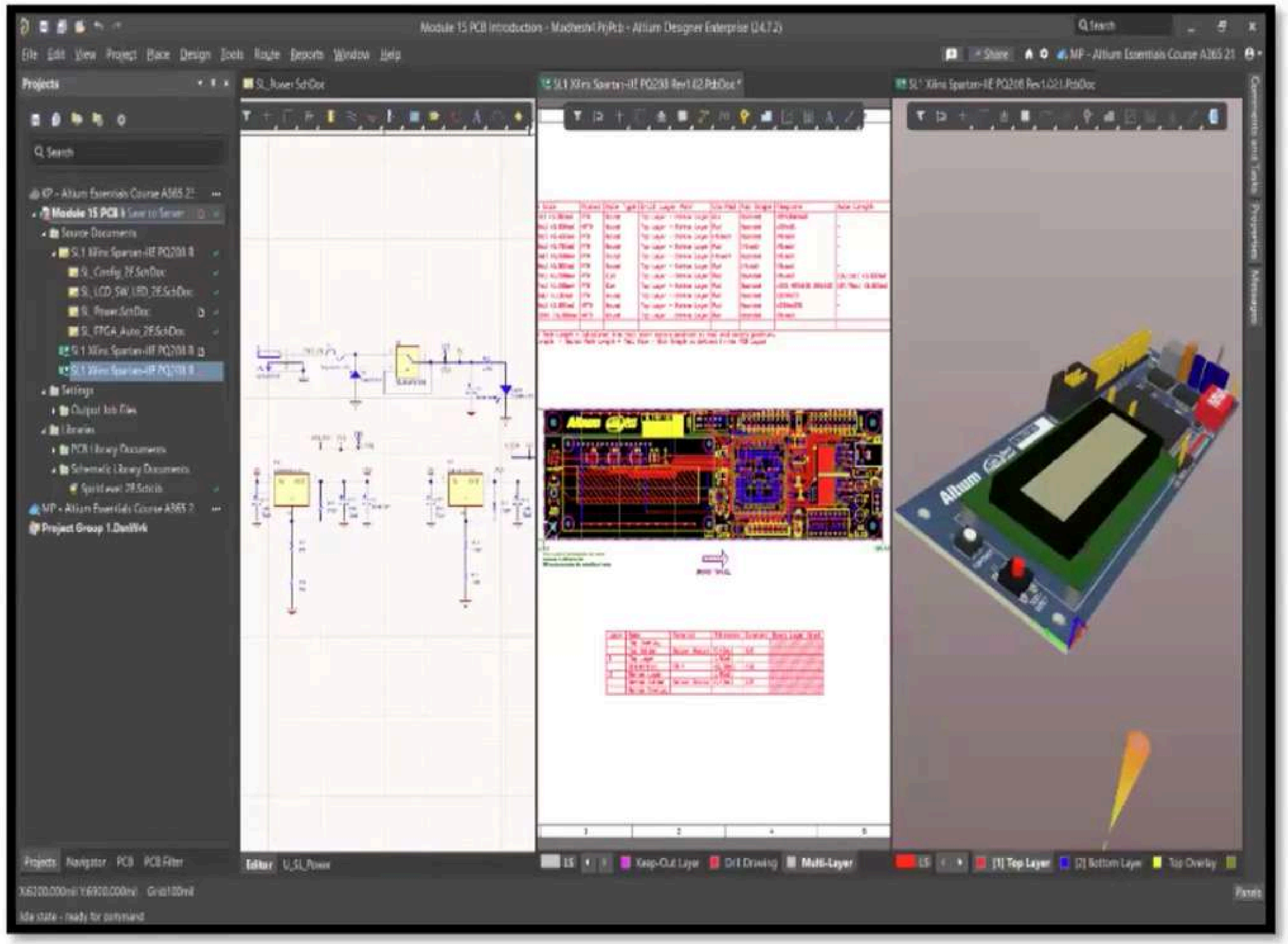
1. Not Scalable
2. Not Reliable
3. Time consuming task
4. Error Prone/Detection is tedious





ECAD- Altium Designer

1. Unified/Integrated Platform
2. Schematic Capturing (ERC)
3. Layout Design (DRC and 3D)
4. Library Creation
5. Simulation / Analysis
6. Fab./Assembly O/P Generation
5. Documentation
5. Data management/Collaboration



Source: Altium Designer (Education)

ECAD Tools in PCB Design

- PCB design is done using specialized ECAD (Electronic CAD) software
- Used for schematic design, PCB layout, and simulation
- Electrical simulators help verify designs before manufacturing
- CAM software prepares design for fabrication and assembly

Goal:

Learn to read and understand PCB layout inside ECAD tools

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Interconnection Parameters

Purpose:

- Design interconnections between PCB components
- Minimize parasitic effects (magnetic & electrical)

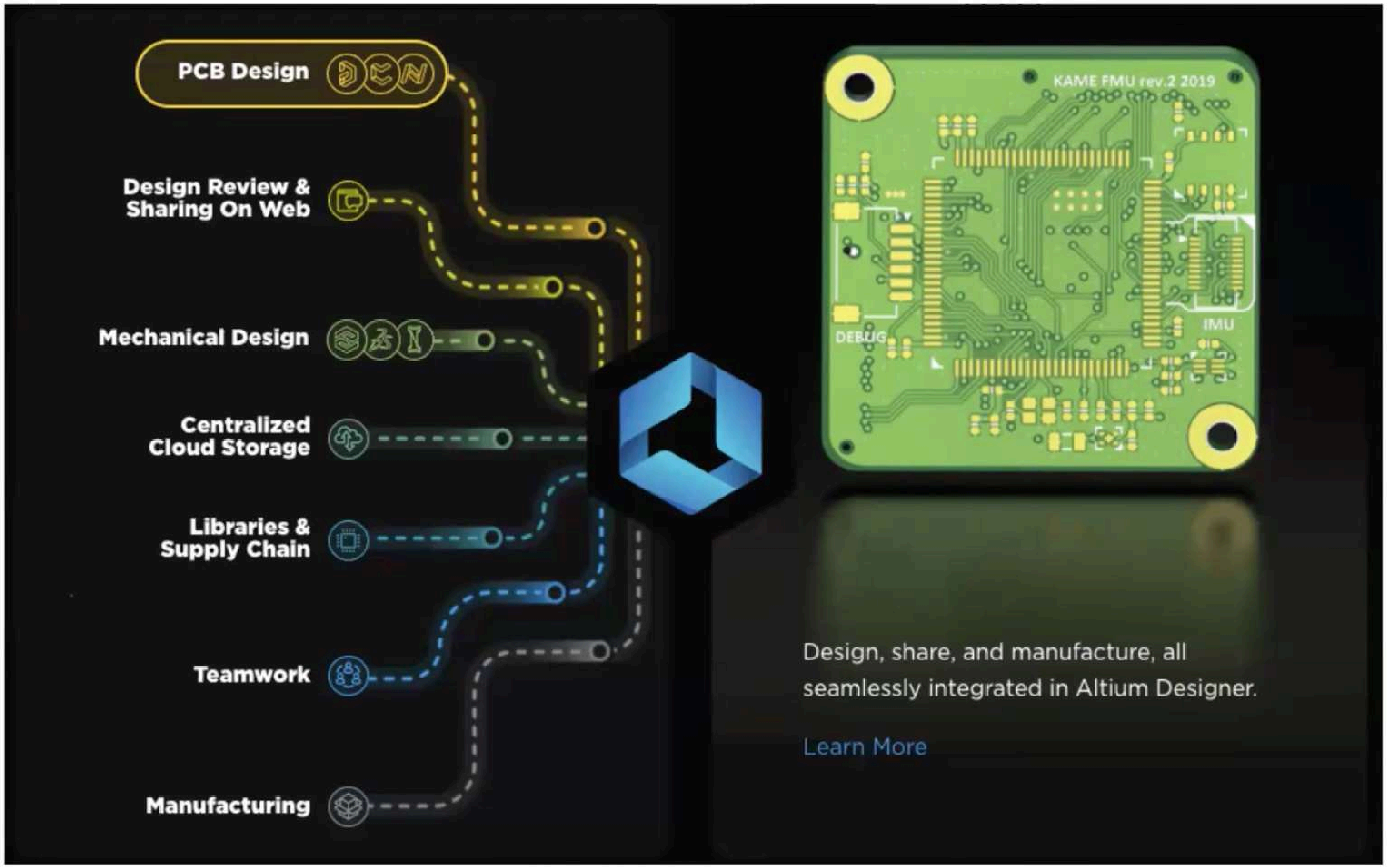
Main Parameters:

- Resistance (R)
- Capacitance (C)
- Inductance (L)

Goal:

Improve signal integrity and circuit performance

Altium 365 Overview



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Day 1 Quiz 6550 1816

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Grid Systems in Layout

- Grid system is widely used in PCB layout design
- Components are placed such that leads align with grid intersection points
- Helps in accurate and uniform component placement
- Simplifies programming of CNC drilling machines
- Improves overall layout organization and readability

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Grid Specifications

- Standard grid unit (for 2:1 scale layout):
- 1 unit = 0.2 inch (≈ 5.08 mm)
- Practical alternative:
- Use 5 mm grid paper for manual layouts
- Typical spacing: 5 mm × 5 mm grid

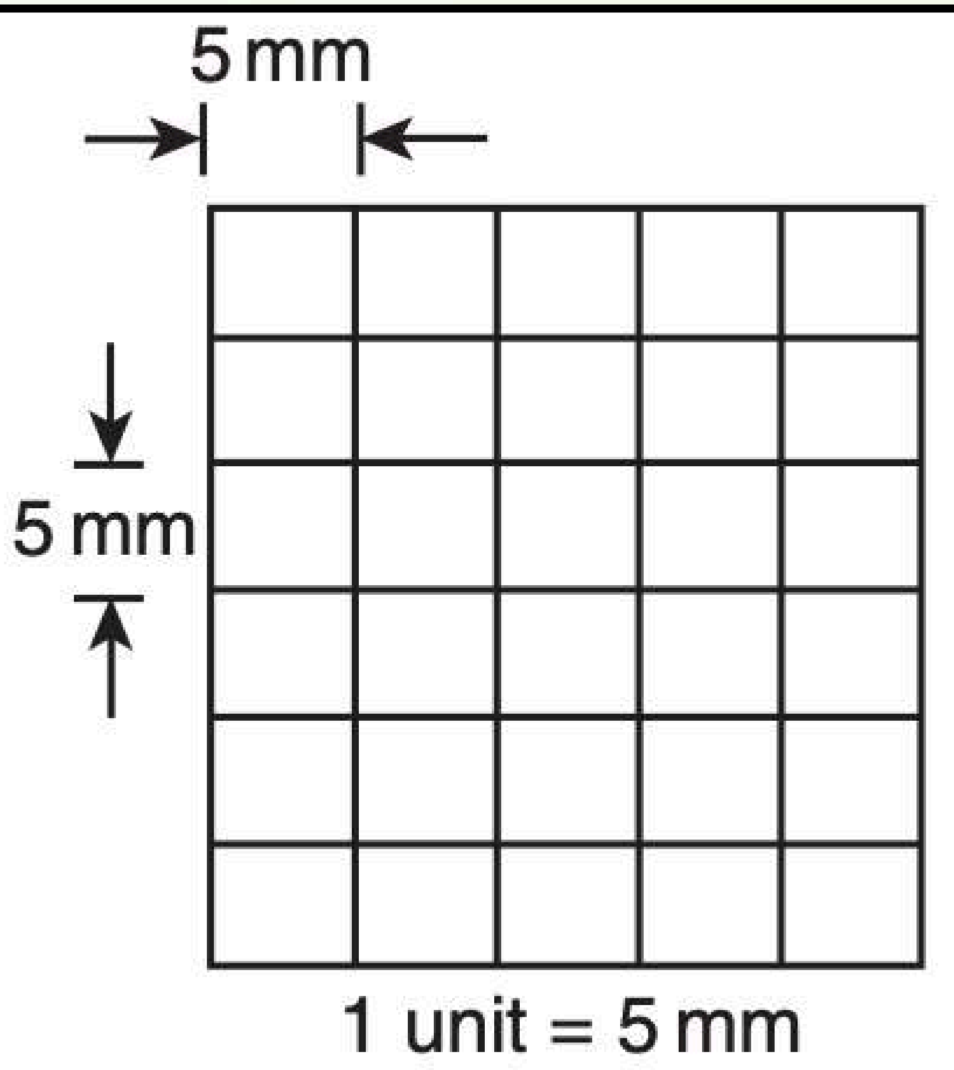

Figure 3.1 A grid system.

Conclusion:

Grid systems ensure precision, uniformity, and ease of PCB layout design

Source: R. G. Kadus – Electronic Product Design

Requirements of Artwork

- Avoid sharp 90° bends (< 0.012 inch spacing)
- Use circular pads/vias
- Pin 1 → square pad
- No T-junctions in signal traces
- T-junctions allowed for power & ground
- Power/GND width ≥ 0.050 inch
- Maintain uniform component orientation

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Silkscreen & Marking

- Show component outline
- Label components (U1, R1, etc.)
- Mark pin 1 / orientation
- Indicate diode polarity
- “+” for polarized capacitors
- Include board name
- Use clear white marking

Placement Guidelines

- Follow board outline
- Number components left → right, top → bottom
- Place decoupling capacitor near each IC

Layout Concepts

- Consider impedance & parasitic effects
- Understand circuit before routing
- Choose trace width based on frequency
- Designer must know circuit + concept

Layout Rules

- Use final schematic only
- Divide into functional blocks
- Follow signal flow direction
- Place large components first
- Fill space with small components
- Keep high-connection parts near connectors
- Ensure easy soldering access
- Use different colors (double layer PCB)

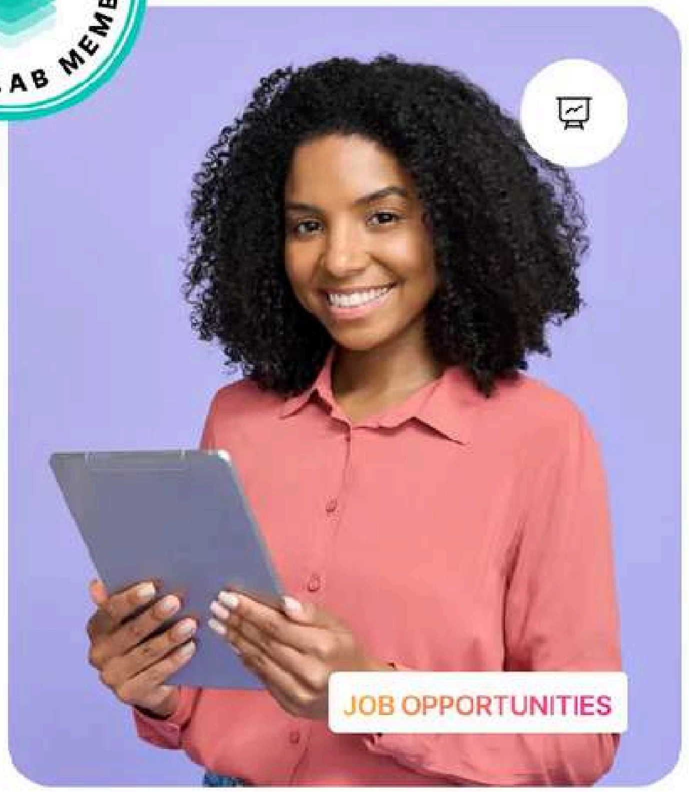
Accelerate Your Career in Electronics Design

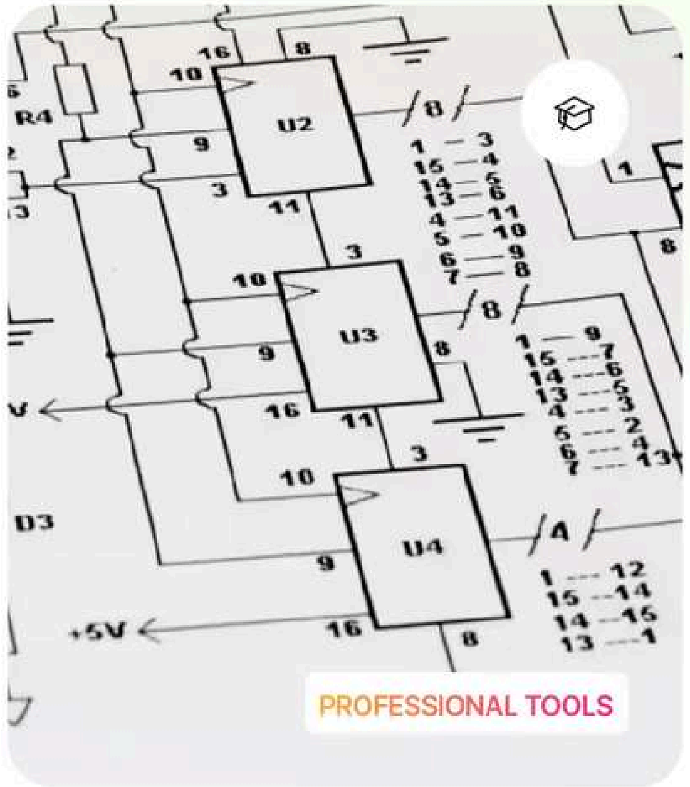
- **Free access to Altium Designer,** the industry's #1 professional PCB design tool.
- **A secure cloud workspace** to collaborate with classmates, track changes, and access projects from anywhere.
- **200,000+ components, templates, and ready-to-use assets** that support real engineering projects.
- **Hands-on, university-level courses** with guided projects to help you master practical PCB design skills.
- **Real-world experience & verified credentials** that make your resume stand out for internships and jobs.

Enroll for Free

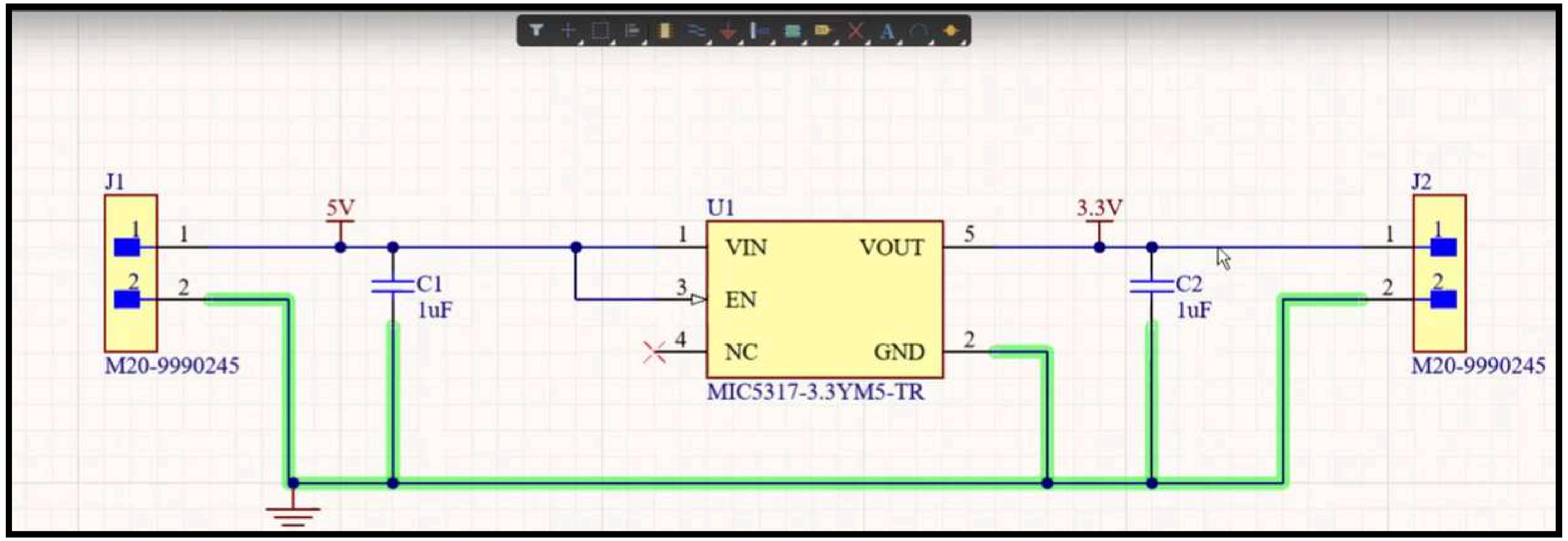
Explore the Lab

Altium EDUCATION STUDENT LAB MEMBER

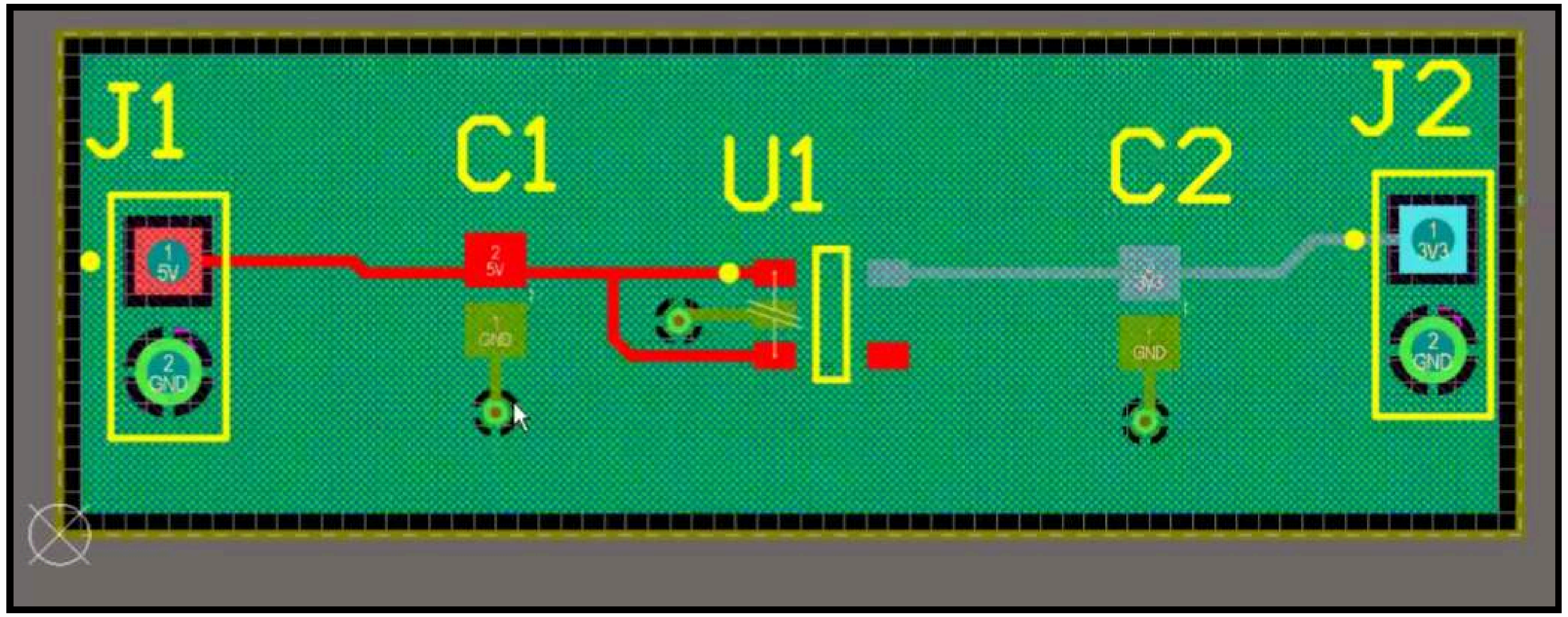




Schematic



J1
1
2
M20-9990245
5V
C1
1uF
U1
VIN
EN
NC
VOUT
GND
MIC5317-3.3YM5-TR
3.3V
C2
1uF
J2
M20-9990245



J1
C1
U1
C2
J2